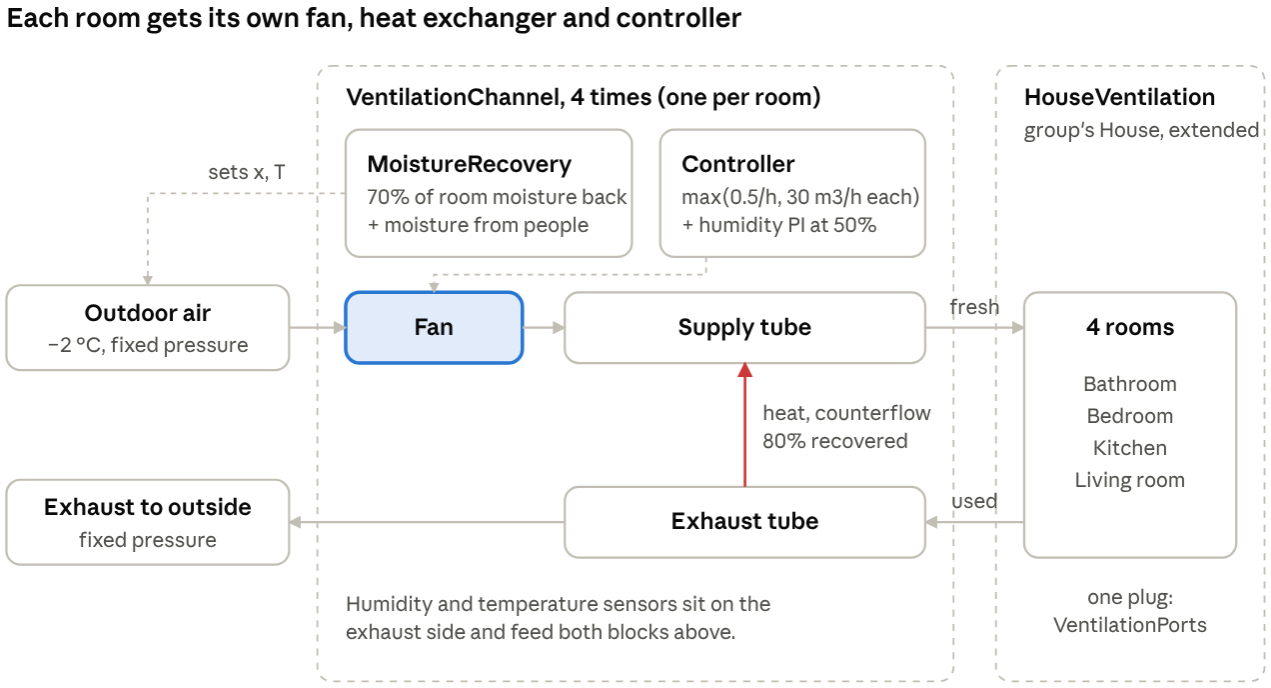
Ventilation model: guide for the report

Oct 30, 2026 · @Pascal

- Scope: ventilation with heat recovery for the Smart House model
- Package: `SmartHouse.ventilation`, branch `ventilation`
- Contents: what was built, why, parameters, results, limits

At a glance



ventilation model · one channel per room

- Fan pushes outdoor air through the supply tube into the room
- Supply tube takes up heat from the exhaust tube (counterflow)
- Used air leaves through the exhaust tube
- Controller sets the fan's airflow
- Moisture recovery sets humidity and temperature of the fresh air
- Solid lines = air, dashed lines = signals

Components

- Ventilated: bathroom, bedroom, kitchen, living room
- Not ventilated: sauna, garage (outside the task)

- Group's files unchanged (House, PartialRoom, Connectors, HouseEnviroment)
- Ventilation attached via `extends` and new models

Model	Type	Does	Why
`VentilationController`	block	airflow = max(0.5 room volumes/h, 30 m^3/h per person) + humidity PI (setpoint 50%)	base and person airflow from sources; PI removes moisture peaks
`MoistureRecovery`	block	fresh-air water content and temperature: 70% of room moisture back + moisture from people	models a rotary (enthalpy) wheel; prevents very dry winter air
`VentilationChannel`	model	one room's ventilation: outdoor boundary, fan, supply tube, exhaust tube, sensors, controller, moisture recovery	reusable, 4 times
`VentilationUnit`	model	4 channels + plug to the house	acts as the central unit
`VentilationPorts`	connector	4 supply + 4 exhaust ports in one plug (orange "air" icon)	1 line instead of 8
`HouseVentilation`	model	group's House via `extends`; ports on in 4 rooms, connected to the plug	no edit of House.mo
`HouseEnviromentVentilation`	model	copy of the group's environment + HouseVentilation + VentilationUnit	model to simulate
`TestHousePhilipp`	model	one-room test	development and checks

Heat recovery principle:

- Supply tube and exhaust tube, 10 cells each

- Coupled cell by cell through heat ports
- Opposite flow directions (counterflow)
- Warm exhaust air heats cold outdoor air

Parameters

- Each value: source or marked assumption
- Fan power not tuned to the Passivhaus value, only compared afterwards

Where	Parameter	Value	Justification
Controller	`ACH_min` base airflow	0.5 room volumes/h	SAENA: minimum 0.5/h
Controller	`personFlow`	30 m³/h per person	SAENA: about 30 m³/h
Controller	`phi_set`	50%	assumption: middle of comfort range
Controller	humidity PI: k, Ti, yMax, yMin	0.0015 (m³/s)/%, 600 s, 100 m³/h, 0	assumption: tuned in one-room test; yMin = 0 → only adds air
Controller	`PhiFilter`	T = 60 s, start 40%	real sensor delay about 1 min; needed for start-up
Unit	room volumes	30.8 / 52.8 / 52.8 / 105.6 m³	House.mo: floor area × 2.2 m
Tubes	geometry	10 pipes, d = 0.05 m, L = 1 m, 10 cells	placeholder for a heat exchanger
Tubes	heat transfer	ConstantAlpha, 100 W/(m²K)	calibrated to about 80% recovery (PHI ≥ 75%, SAENA 80–90%)
Tubes	pressure drop	QuadraticMassFlowDependent, 100 Pa per tube at 0.02 kg/s	PHI test pressure 100 Pa; design point 2 people = 60 m³/h; ducts + filter + exchanger
Fan	`SimpleFan`, airflow preset	eta = 0.33, etaDrive = 1	REHVA: measured average total efficiency 33%; motor losses included

Where	Parameter	Value	Justification
MoistureRecovery	`eta_x`	0.7	assumption: typical rotary wheel; 0 = plate exchanger
MoistureRecovery	`phi_out`	80%	assumption: typical winter value
MoistureRecovery	`moisturePerPerson`	90 g/h	typical at rest (ASHRAE), not re-checked
MoistureRecovery	`h_evap`, `cp_air`	$2.5{\cdot}10^6$ J/kg, 1006 J/(kg K)	physical constants
Rooms	`phiInitial`	40%	away from setpoint, for start-up
Rooms	`fixedInitialPressure`	false	outdoor boundaries fix the pressure
Environment	outdoor temperature	−2 °C (271.15 K)	group's HouseEnviroment

Results of the base case

- Fans: about 13 W mean
- Heat recovered: about 810 W
- Fans = about 1.6% of recovered heat
- Run: 1 day, −2 °C outdoor, group's occupancy (2 people per room, 2–4 h)

Room	Mean airflow	With 2 people	Max airflow	Fan mean / max	Fan energy	Heat recovered (mean)	Efficiency (12 h)	Humidity min–max
Bathroom	19 m³/h	60 m³/h	60 m³/h	1.1 / 11 W	26 Wh/day	133 W	85%	8.8–52.2%
Bedroom	30 m³/h	60 m³/h	60 m³/h	1.9 / 11 W	45 Wh/day	181 W	82%	13.3–52.4%
Kitchen	30 m³/h	60 m³/h	60 m³/h	1.9 / 11 W	46 Wh/day	191 W	82%	11.5–52.8%
Living room	54 m³/h	60 m³/h	82 m³/h	8.6 / 28 W	205 Wh/day	309 W	75%	14.3–52.6%
House	133 m³/h			**13.4 W**	**0.32 kWh/day**	**814 W (19.5 kWh/day)**		

Validation against sources:

- Heat recovery 75–85%: within PHI ⩾ 75% and SAENA 80–90%
- Lowest in living room: most air through the same exchanger
- Fan at design point: 60 m³/h, 200 Pa → about 11 W
- Hand check: $P = \dot{V} \cdot \Delta p / \eta = 0.0167 \times 200 / 0.33 \approx 10$ W
- 0.18 W per m³/h: below Passivhaus limit 0.45, near certified unit 0.30
- Fan average 0.10 W per m³/h: rooms mostly below design point, $\Delta p \sim \text{airflow}^2$
- Supply air 17.7–25 °C: no cold draught
- Mass balance: supply = exhaust within 0.3%

Reading the 814 W:

- Measured against air entering the exchanger
- That air is pre-warmed by the moisture recovery (see Limitations) → true saving lower
- Exact saving: heating energy with vs. without heat recovery

Modelling decisions

- Each decision needs one line of justification in the report

Decision	Why	Alternative not taken
One channel per room, not one central unit	controller per room = ideal damper (VAV valve); same fan power and recovery	central unit with dampers, mixed exhaust: more work, little change
Exchanger = two counterflow tubes, ConstantAlpha calibrated to about 80%	placeholder geometry → physical correlations wrong: Gnielinski about 2 W/(m²K) (recovery under 10%), Konakov about 0.2 Pa	real plate-exchanger geometry: not possible with the same working fluids
One fan per room on supply side; both boundaries at fixed pressure	one component sets the airflow; two flow-forcing fans → room pressure drifts	pressure-controlled exhaust fan
Fan pressure covers both tubes (about 200 Pa at 60 m³/h)	room in slight overpressure pushes air out	**do not double the fan power**
Pressure drop by nominal values	ducts + filter + exchanger together; 1 m tube alone about 1 Pa	duct friction from geometry

Decision	Why	Alternative not taken
Moisture recovery as own block → outdoor boundary	TIL tubes exchange heat, not moisture	real enthalpy-exchanger model
Fresh-air temperature + latent heat of recovered water	otherwise fog at the cold boundary --> moisture exchanger doesnt do it in reality	water added after the exchanger: not possible here
Humidity filter (FirstOrder, 60 s) before the PI	fixes start-up (otherwise P control start problems) , behaves like a real sensor	start values only: unreliable

Limitations the report must state

- Rows 1–2: errors in our own model
- Rest: simplifications

Limitation	Evidence	Effect
Moisture recovery adds heat that does not exist	fresh air pre-warmed from −2 °C up to 14.8 °C; about 20% reaches the rooms	about +50 W into rooms (about 6% of recovered heat); 814 W too high
No background moisture	only people release moisture; real homes 2–9 l/day (SAENA)	humidity 9–14% in empty rooms (too low)
Rooms in overpressure	living room +128 Pa, others +11 to +32 Pa; real: a few Pa	none (rooms airtight)
Constant fan efficiency	real fans worse at part load, standby power	fan energy 0.32 kWh/day = lower bound
No condensate or frost model	exhaust saturated at exchanger outlet 1.6–4.1 h/day; exhaust out ≥ 3.5 °C	valid for mild winter; below about −10 °C frost protection needed
No shower moisture	BathroomPlaceholder has no shower	humidity PI hardly used
One exchanger per room	real central unit mixes all exhaust air	small

Sources

- Passivhaus Institut, certificate residential unit EcoVent 400: limit ⩽ 0.45 Wh/m³, recovery ⩾ 75%; unit 0.30 Wh/m³, 85%; tested at 100 Pa
- SAENA, brochure Wohnungslüftung: 0.5/h (ch. 2), about 30 m³/h per person (ch. 1.4), 2–9 l moisture/day (ch. 1.3), recovery 80–90% (ch. 4.1)
- N. Brelih, REHVA Journal, Feb. 2012: average total fan efficiency 33% (767 fans), residential lowest
- Regulation (EU) No 327/2011: fan rules only from 125 W → no minimum for our 2–30 W fans
- Wikipedia, Specific fan power: SFP = P / V̇
- DIN 1946-6 (home ventilation): main German standard, not available online, not cited